

Analyzing an organism's sensors using Maximum Entropy models with bias, variance, and confusion matrices

Christopher Wang¹ Elianna Schimke¹ Tristan Kako^{2,3} Aiden Gao³ Martina Lamberti^{4,5} Joost le Feber⁵ Sarah Marzen^{1,6,7}

¹Department of Natural Sciences, Scripps and Pitzer College

²Tufts University, MA

³Pioneer Academics, PA

⁴Center for Research in Neuroscience of LYON, Bron, France

⁵University of Twente, The Netherlands

⁶Claremont McKenna College

⁷National Institute for Theory and Mathematics in Biology, IL

Introduction

All living organisms use external stimuli to navigate their environment via their sensors. Because encoding information costs energy, organisms retain only a fraction of the information received from their sensors, ideally information that maximizes their ability to remember past environmental states or predict future ones, key functions that support survival. To better understand how well sensor systems absorb stimulus information, we used stimulus-dependent Maximum Entropy (MaxEnt) models with maximum likelihood estimation and typical statistical metrics, such as confusion matrices or bias and variance. In this poster, we apply MaxEnt to investigate cooperativity (using bias and variance) and memory/prediction in neural systems (using confusion matrices). This approach provides two primary benefits over previous approaches: it is more computationally efficient, and it provides a more information-rich picture on how sensors interact with stimulus.

Materials and Methods

Statistical mechanical models serve as stimulus-dependent Maximum Entropy (MaxEnt) models, yielding $p_{\text{model}}(s | x)$, where s is the sensor state and x is the environmental variable. Here we use Bayes' theorem,

$$p(x | s) = p(s | x)p(x)/p(s), \quad (1)$$

and assuming a uniform prior $p(x)$, we form a Maximum Likelihood Estimator (MLE) $\hat{x}(s) = \arg \max_x p_{\text{model}}(s | x)$.

Estimator Construction for Ordered Variables

With the MLE in hand, we evaluate the estimator's quality as a function of the true environmental state via

$$\text{Bias}(x) = \langle \hat{x}(s) \rangle_{p_{\text{model}}(s|x)} - x \quad (2)$$

$$\text{Var}(x) = \langle (\hat{x}(s) - \langle \hat{x}(s) \rangle_{p_{\text{model}}(s|x)})^2 \rangle_{p_{\text{model}}(s|x)} \quad (3)$$

$$\text{MSE}(x) = \langle (\hat{x}(s) - x)^2 \rangle_{p_{\text{model}}(s|x)}. \quad (4)$$

To investigate cooperativity—a case where we draw new insights using MaxEnt—we use a two-site Adair model in which two binding sites s_1, s_2 interact with ligand concentration x through a partition function

$$Z = 1 + 2(x/k) + e^{-\beta\Delta\epsilon}(x/k)^2, \quad (5)$$

where K is the dissociation constant and $\beta\Delta\epsilon$ is a cooperativity parameter: negative values correspond to positive cooperativity (binding at one site favors binding at the other) and positive values to negative cooperativity.

Neural system experimental setup

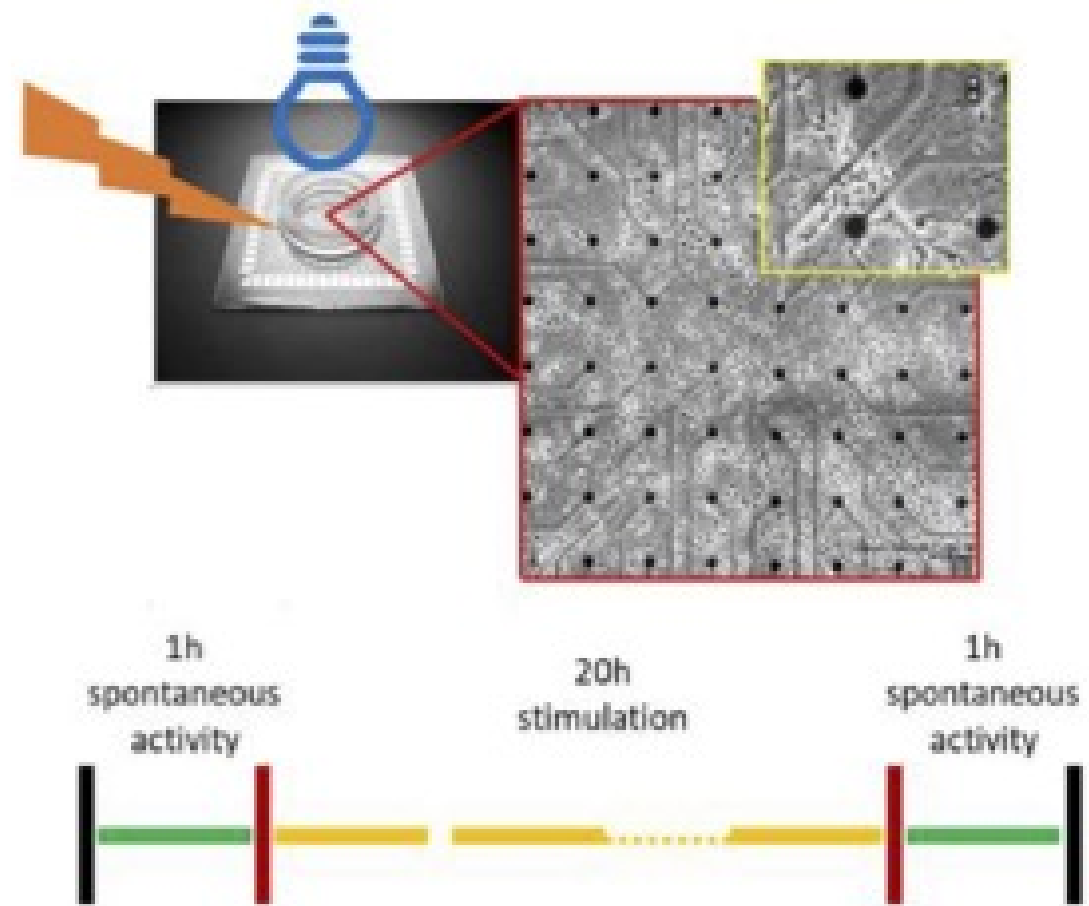


Figure 1. Example of MEA with cultured neuronal networks and stimulation protocol.

Newborn rat neurons were placed on multi-electrode arrays (MEAs) and incubated for 3 weeks in the following conditions: 36°C, high humidity, and 5% CO₂. 59 electrodes were connected to the system to measure activity at a sampling frequency of 16 kHz. We recorded 20 hours of light (global) or electrical (focal) stimulation with 1 hour of spontaneous activity on either side. Focal electrical stimulation was applied through one single electrode and global (optogenetic) stimulation used LED pulses placed above the MEA.

Data Processing and Model Fitting for the Neural System

Our estimator guesses stimulus in/activity (categorical variable), so it is most productive to characterize model performance by generating confusion matrices against the true stimulus. The confusion matrix tells us how often we correctly guess stimulus activity (true positive rate, TPR), stimulus inactivity (true negative rate, TNR), and how often we mistake one for the other. From this, we can derive a predictive accuracy, indicating how often our model makes a correct guess.

Materials and Methods (continued)

Neural activity and stimulus were binned into successive windows of width $\Delta t = 100$ ms, where one or more spikes in a window yields a 1 and no spikes yields a 0, producing a binary matrix. We fit a joint MaxEnt spin-glass Ising model to estimate the coupling matrix J via

$$p_{\text{model}}(x, s) \propto \exp \left(- (x \ s) \underbrace{\begin{pmatrix} J_{xx} & J_{xs} \\ J_{sx} & J_{ss} \end{pmatrix}}_J \begin{pmatrix} x \\ s \end{pmatrix} \right), \quad (6)$$

whose leading entry J_{xx} captures the firing propensity of the stimulus, off-diagonal blocks J_{xs} and J_{sx} encode stimulus–network correlations, and whose diagonal block J_{ss} captures network–network interactions. The stimulus channel was time-shifted relative to network activity by up to ± 200 ms to probe memory (negative shifts) and prediction (positive shifts). The fitted J matrix was then used to compute the likelihood ratio

$$L(s) = \frac{p(x=1 | s)}{p(x=0 | s)} = \exp \left(J_{xx} + J_{xs}s + s^\top J_{sx} \right). \quad (7)$$

If $L(s) > 1$, $\hat{x}(s) = 1$, otherwise $\hat{x}(s) = 0$.

Results and Discussion

Cooperativity in Sensors

Fig 2 shows bias, variance, and MSE for the two-site Adair model across a range of cooperativities $\beta\Delta\epsilon$. Cooperativity reshapes which ligand concentrations the sensor estimates well: there exists a range of concentrations for which positive cooperativity lowers MSE relative to the independent case ($\beta\Delta\epsilon = 0$), and a complementary range for which it does not. This stands in contrast to a standard mutual information (MI) analysis, which finds that conditionally independent sensors always transmit more total information. Our analysis reveals that the answer to whether cooperativity benefits sensing depends on which ligand concentrations the organism is likely to encounter, a nuance invisible to MI alone.

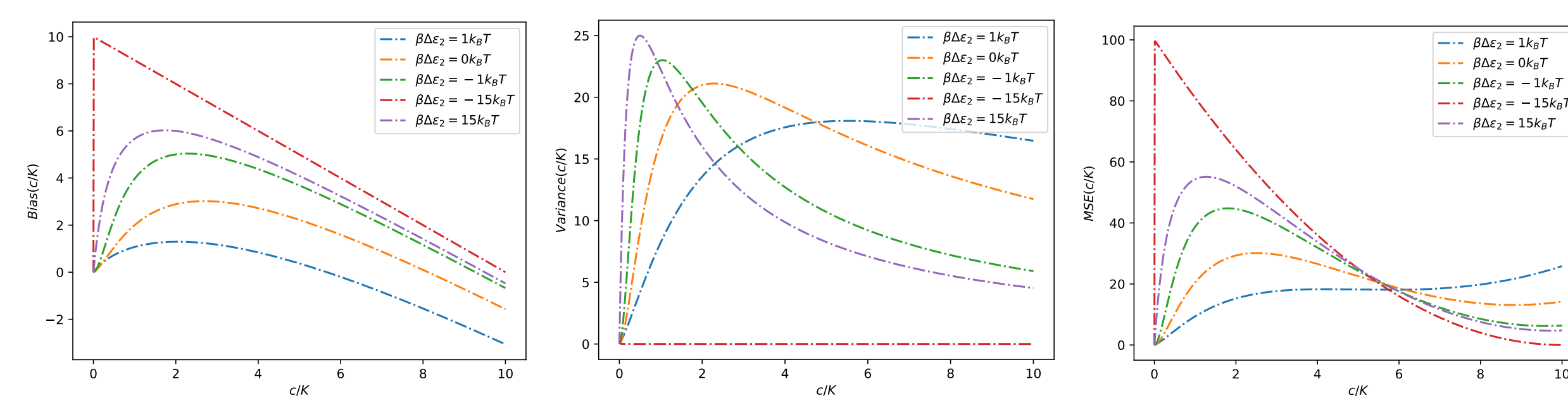


Figure 2. The bias, variance, and MSE of a sensor with varying degrees of cooperativity. At left, bias as a function of the chemoattractant concentration c . In the middle, variance as a function of the chemoattractant concentration c . At right, mean-squared error (MSE) as a function of the chemoattractant concentration c . A range of cooperativities based on varying $\beta\Delta\epsilon$ are shown.

Memory and Prediction in Cultured Neurons

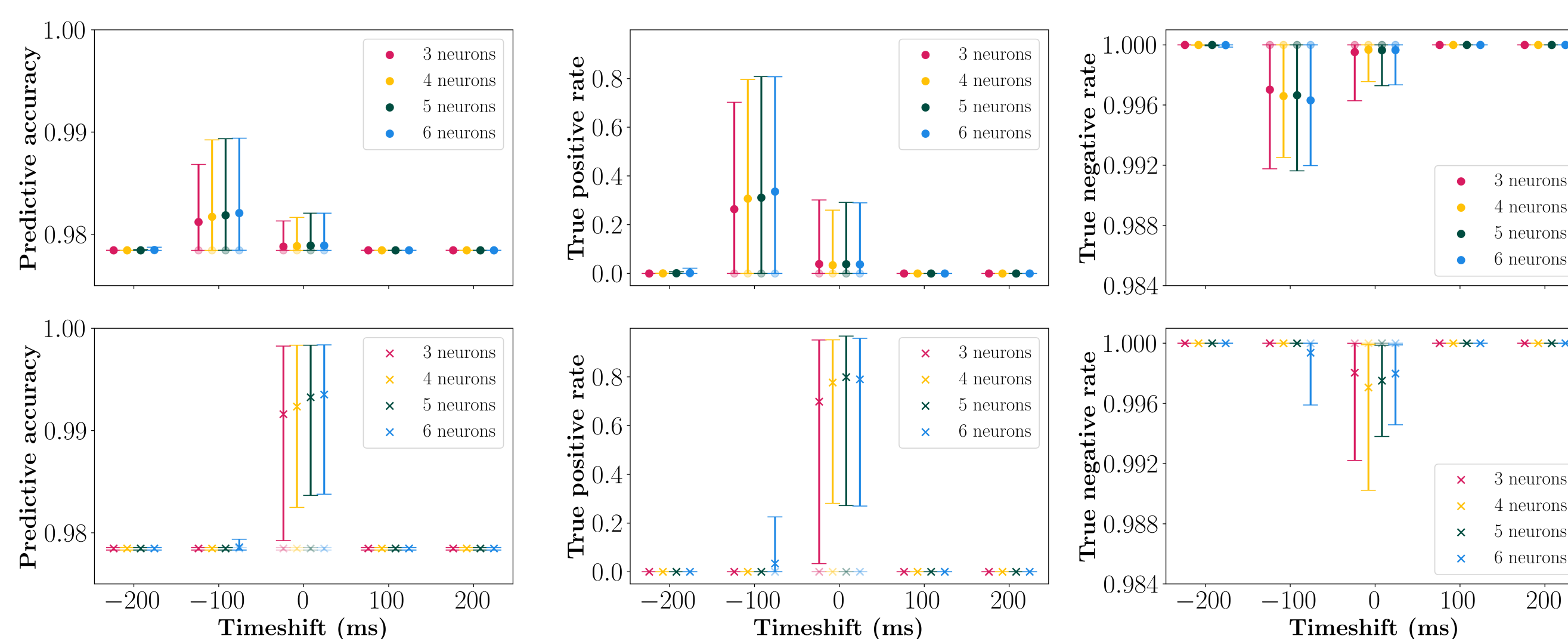


Figure 3. Predictive accuracy, true positive rate, and true negative rate for baseline and model estimators with 95% confidence intervals. Circular markers denote optogenetic ($n = 10$) stimulation (top row), crossed markers denote electrical ($n = 8$) stimulation (bottom row). Plots with varying system sizes at the same time-shift are staggered to enhance visibility of markers and error bars.

Results and Discussion (continued)

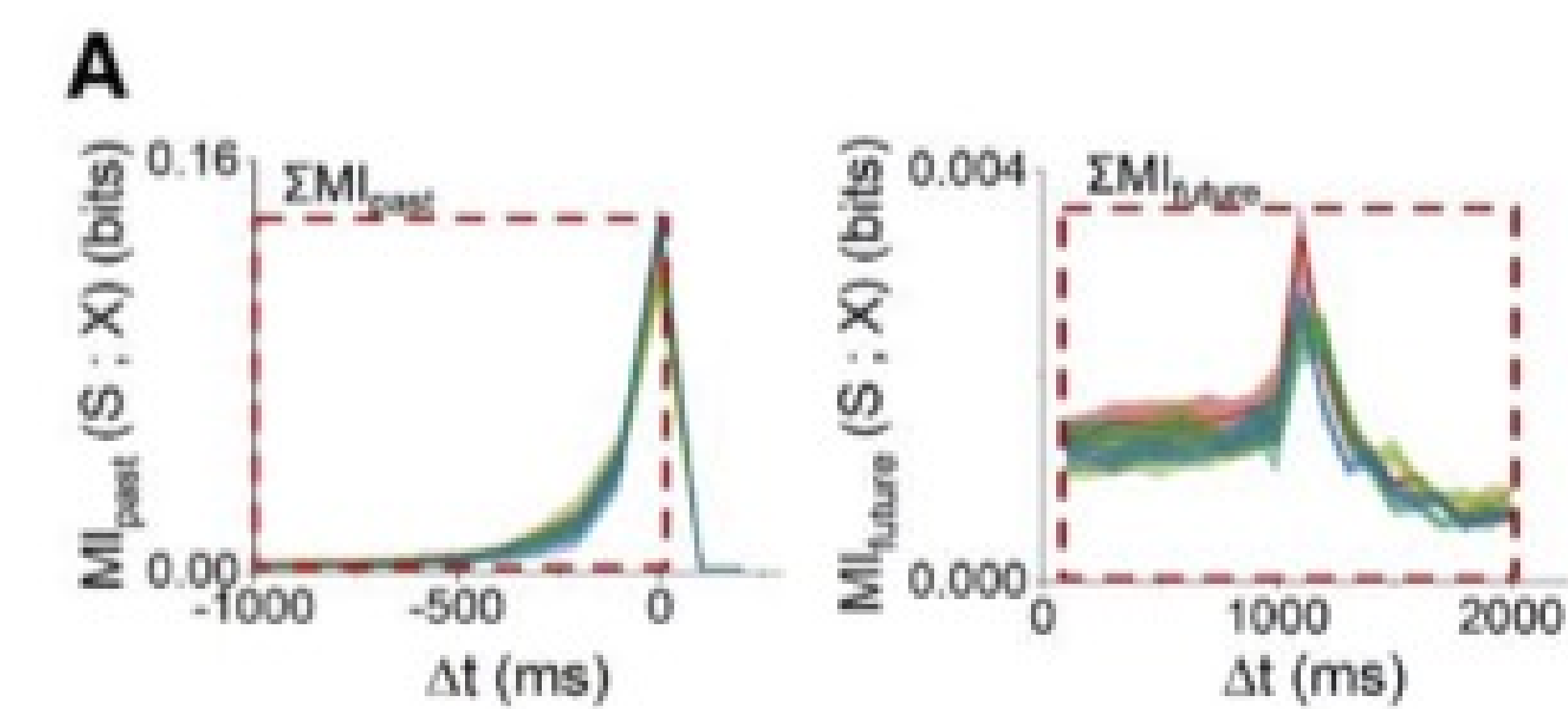


Figure 4. An example of MI_{past} and MI_{future} . Taken from Fig 3A of [3].

Fig 3 shows that electrical stimulation yields the strongest improvement over baseline at 0 ms time-shift (mean predictive accuracy 99.35%, TPR 79.05%), while optogenetic stimulation peaks at -100 ms (predictive accuracy 98.21%, TPR 33.64%). Even selecting just the three most informative electrodes captures the majority of possible improvement, with diminishing returns at larger subsystem sizes. Notably, no system displayed predictive behavior: at time-shifts of $+100$ ms and beyond, all metrics returned to baseline. This is consistent with MI analyses, which found $MI_{\text{future}} \sim 10^{-3}$ bits (mutual information of prediction) compared to $MI_{\text{past}} \sim 10^{-1}$ bits (mutual information of memory). The predictive signal is likely too weak to cross the likelihood ratio threshold, while the memory signal is strong enough to drive improvements. Moreover, confusion matrices add granularity: it separately quantifies how often the network mistakes silence for activity versus the reverse.

We also experimented with changing time bin size and decomposing each experiment into hourly chunks, but found no statistically significant trends.

Conclusion

We combined stimulus-dependent and joint MaxEnt models with maximum likelihood estimation to analyze biosensors using bias, variance, and confusion matrices, metrics that provide a richer picture than MI alone. For ordinal variables, bias and variance as functions of ligand concentration revealed that cooperativity can improve estimation at specific concentration ranges, a trend invisible to MI. For categorical variables, confusion matrices quantified memory and prediction in cultured neurons, showing strong memory signatures consistent with prior MI analyses while additionally resolving error types across activity and inactivity.

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